

A Comparative Study of Fine-Tuned vs. Zero-Shot LLMs for Organizational Role & Business Unit Prediction

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1. Introduction and Motivation

In the past, predicting an employee’s organizational role or business unit has been approached as a network problem. However, with the rapid advancement of Natural Language Processing, text-understanding models have become the default tools for enterprise data classification. This project isolates the text from the network entirely to ask a more relevant modern question: How do different generations of transformer architectures compare when predicting an employee's organizational standing purely from the semantic linguistic content of their emails?

Specifically, we evaluate the performance trade-offs between two distinct paradigms: specialized, lightweight fine-tuning (DistilBERT and RoBERTa) versus the out-of-the-box zero-shot capabilities of a massive foundational model (Gemini 3 Flash). By strictly isolating the text, we aim to discover whether massive pre-training can outperform domain-specific fine-tuning when interpreting highly specialised corporate jargon.

2. Related Work

As mentioned in the introduction, existing literature on predicting roles and business units using the Enron dataset relies on network structures. Linguistic research has connected specific phrasing to upward or downward corporate communication and combining these semantic insights with modern transformer architectures offers a novel methodology.

3. Methodology

To ensure our models learned true role-predictive language rather than memorizing employee names or signatures, we implemented a strict anonymization pipeline and employee-level data splitting.

Employee Level Splitting - If the same employee appears in both the training and testing sets, the model could trivially memorize their name or writing style (things like most likely phrases). Thus, we split our data at the employee level so that 80% of the employees are assigned to the training set, 10% to validation, and 10% to testing. This forces the model to generalize to entirely unseen individuals.

Data Sanitization Pipeline - We applied several layers of sanitization to prevent data leakage. First, all email headers were removed, including From:, To:, CC:, BCC:, etc. Then, we utilized spaCy’s Named Entity Recognition to mask the names of senders and coworkers in the email body with a [PERSON] token. Finally, we removed signature blocks, which sometimes leak data like job titles and phone numbers.

4. Experimental Setup & Baselines

To evaluate trade-offs between model size and performance on fine-tuned tasks, we utilized 2 distinct architectures: DistilBERT for granular Role identification and RoBERTa for broader Business Unit prediction.

To properly contextualize the performance of these classifiers, we established a zero-shot baseline using Gemini 3 Flash. By comparing specialised fine tuning against zero-shot capabilities of a state-of-the-art general LLM, we gain insight into how inherently difficult it is to complete this classification task. By demonstrating how non-trivial it is to deduce organizational standing from text alone, this baseline ensures that our models' performance provably validates our core hypothesis: semantic linguistic insights are a highly effective, standalone predictor of corporate roles and business units.

5. Results & Empirical Analysis

Our evaluation yielded highly successful results that validate the use of text-only classifiers for organizational prediction tasks, though performance varied significantly depending on the granularity of the target labels.

Business Unit Prediction

On the Business Unit prediction task, our fine-tuned RoBERTa model achieved an outstanding 96.1% Accuracy and 0.916 Macro F1. This comprehensively proves our research hypothesis which aimed to test whether departments within an organization (e.g., Legal Department vs. Regulatory Affairs) utilize highly distinct semantic vocabularies that a fine-tuned transformer can reliably map without any reliance on traditional graph features.

Notably, RoBERTa significantly outperformed the zero-shot baseline, which achieved 73.9% Accuracy. The zero-shot baseline revealed a critical brittleness associated with out-of-the-box LLMs: strict formatting compliance. This was discovered as we noticed the Enron data set contains a misspelled ground-truth label and Gemini kept trying to autocorrect the spelling in its output which caused a 0.0 F1-score for that class. RoBERTa mapped features regardless of typos and showcased robustness when dealing with noisy, real-world data.

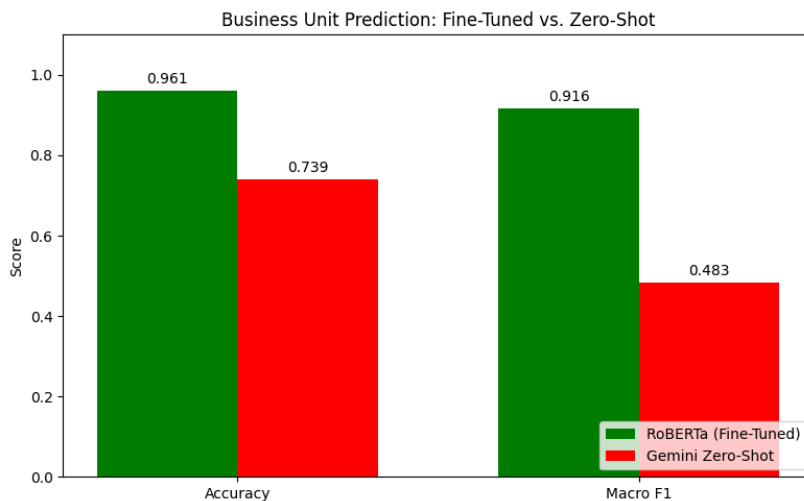


Figure 1. Graph depicting the comparison between the Accuracy and Macro F1 scores of RoBERTa and Gemini Zero-Shot model for the task of business unit prediction

Role Prediction

When evaluating the more granular task of hierarchical Role prediction, our analysis revealed an interesting trade-off. Our fine-tuned DistilBERT model achieved a higher overall accuracy (62.7%) than the zero-shot baseline (56.0%). However, DistilBERT's Macro F1 score (0.160) highlights the limitations of lightweight models on heavily skewed datasets. A review of the DistilBERT confusion matrix shows that instead of learning the distinguishing semantic features of minority classes, the lightweight architecture collapsed into predicting the two dominant majority classes: 'Vice President' and 'Employee'.

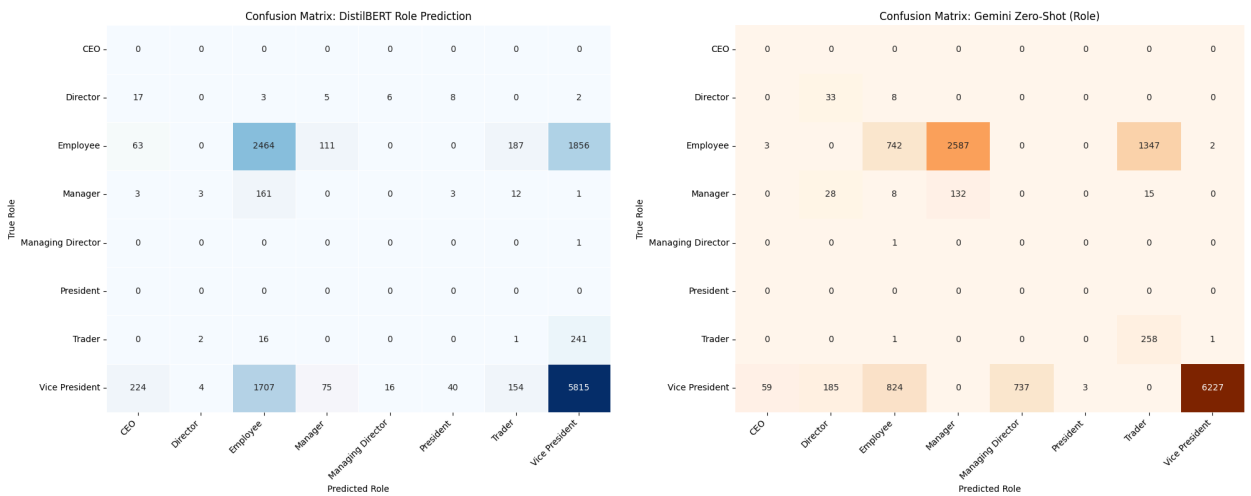


Figure 2. Confusion matrices of DistilBERT’s and Gemini Flash 3’s role prediction.

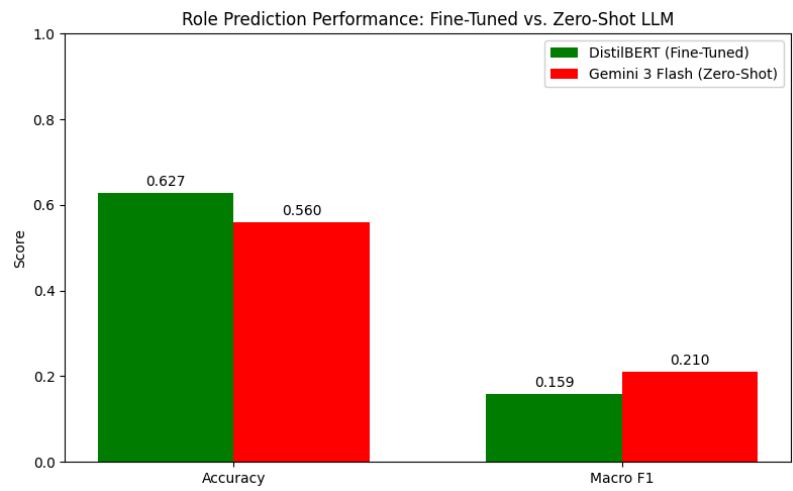


Figure 3. Graph depicting the comparison between the Accuracy and Macro F1 scores of DistilBERT and Gemini Zero-Shot model for the task of role prediction

In contrast, the zero-shot baseline achieved a better Macro F1 score (0.213). By comparing the confusion matrices we can see that this is the case due to DistilBERT’s vast pre-training. The zero-shot model was able to detect the nuanced semantic markers of specialized

roles due to its vast general knowledge. It achieved a 99% recall for Traders and an 80% recall for Directors which were classes that DistilBERT entirely misclassified. This suggests that using lightweight models for departmental classification and foundational models for niche hierarchical roles can yield an optimal text-only performance.

7. Conclusion

This study set out to determine if lightweight, fine-tuned LLMs could predict organizational roles purely from linguistic content, offering a viable alternative to traditional graph-based network features. Our results indicate that text-only classifiers are remarkably effective. As noted, fine-tuned models like RoBERTa can map semantic text to broad Business Units with near-perfect precision (96% accuracy).

At the same time, our baseline comparison proved that foundational models possess a superior ability to detect semantic cues for niche corporate roles compared to smaller, domain-specialized transformers facing class imbalances. Ultimately, it can be concluded that linguistic content alone contains organizational signals that are highly predictable. Thus proving that modern text-understanding architectures can successfully map corporate power dynamics without relying on network metadata.

8. Summary of Contributions

Veer Nishchal Guda - Built project plan for midterm proposal. Worked on the linguistic feature extraction and interpretability pipeline that exists in `explain.py`. Implemented the TF-IDF vectorization and Chi-squared statistical testing to map n-grams and phrases to corporate roles which was crucial to help models train. Also built the preprocessing pipeline in `preprocess.py` and the training architecture present in `train.py`.

Pavan Kizhakkemadham Ananthakrishnan - Coordinated efforts and led discussions. Worked on writing out the motivation for the project proposal. Developed the evaluation pipeline in `evaluate.py` which calculates multi-class performance metrics like Macro/Micro F1 & Accuracy and generates confusion matrices. Wrote out the notebook file to tie all of our individual work together so that it can be run on Google Colab.

Laksh Makhija - Worked on finding the initial dataset and helped build the research question. Made adjustments to our final project approach based on feedback given during the project proposal stage. Worked on the inference script present in `predict.py`. Wrote out the final project document after gathering results and making conclusions.